Validating a Bitcoin Forecasting Model: Look-Ahead Bias in VMD-Based Features

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Abstract

A multi-horizon CNN-BiLSTM model forecasting hourly Bitcoin returns from a Variational Mode Decomposition (VMD) of past returns appeared to predict the direction of the next hours correctly about 8 times out of 10. We show that this apparent skill is entirely due to look-ahead bias: when the VMD is applied to a whole block of data, each mode contains information from future prices. Once the decomposition is restricted to past data only, the same network, trained on the same data, does not outperform a random walk over a 15-month out-of-sample period.

1 Model and data

Data. Hourly BTC/USDT closing prices from Binance, January 2018 to October 2026 (about 77,000 observations), with hourly log-returns $r_t = \ln P_t - \ln P_{t-1}$.

Features. The VMD splits a signal into $K = 3$ band-limited modes (penalty $\alpha = 2000$). The network receives the last 24 hourly values of the three modes.

Targets. Log-returns between successive horizons $h \in \{1, 2, 4, 6, 8, 12, 24\}$ hours, chained to obtain a forecast path.

Network. Two 1-D convolution layers (32 and 64 filters) followed by a two-layer bidirectional LSTM (64 units) and a linear output layer; Adam optimiser, mean squared error, early stopping on the validation set.

Chronological split. Training: January 2018 – June 2024; validation (early stopping): July 2024 – June 2025; test: July 2025 – October 2026 (11,240 hourly forecasts), never used for training or model selection. A 24-hour gap separates the periods so that targets do not overlap.

2 Two ways of computing the features

- **VMD on whole blocks** (original approach): the decomposition is run once on a long series (here, one calendar year at a time). The value of a mode at hour t then depends on the returns observed *after* t.
- **Causal VMD** (corrected approach): for each forecast time t, the decomposition is run only on the 256 returns ending at t. We checked that the features of a sample are strictly identical when all data after t is deleted.

3 Out-of-sample results

Accuracy is compared with a random walk (forecast of zero return), using the out-of-sample $R^2 = 1 - \sum(y - \hat{y})^2 / \sum y^2$ on cumulative log-returns, and the share of forecasts with the correct sign.

	Causal VMD		VMD on whole blocks	
Horizon	R^2 vs random walk	Direction correct	R^2 vs random walk	Direction correct
1h	-0.0005	50.3%	+0.62	75.8%
2h	-0.0021	50.6%	+0.71	79.5%
4h	-0.0021	49.9%	+0.62	76.5%
6h	-0.0023	49.3%	+0.42	71.4%
8h	-0.0020	50.6%	+0.21	66.6%
12h	-0.0007	51.1%	+0.00	58.6%
24h	-0.0006	51.5%	-0.05	56.2%

Table 1: Test period July 2025 – October 2026. Same network, data and split; only the feature computation differs.

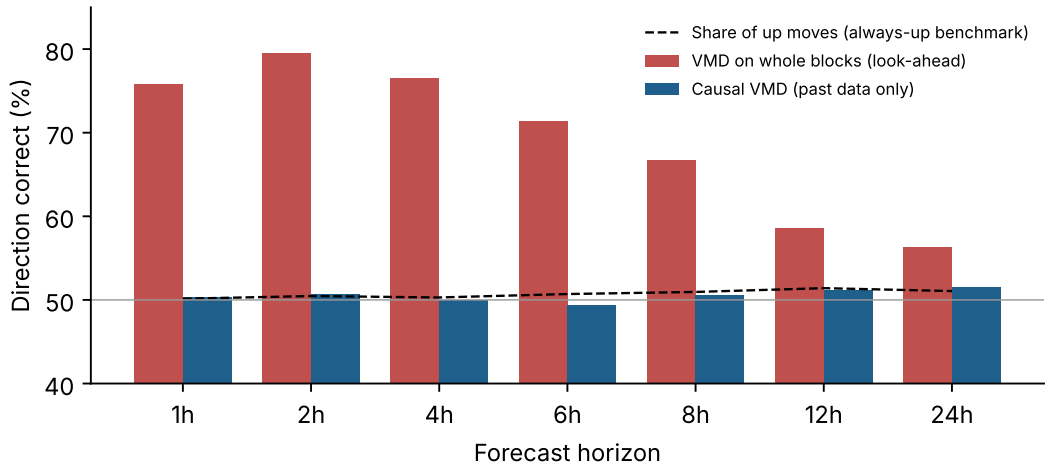


Figure 1: Share of correct directional forecasts. The dashed line is the share of up moves, i.e. the score of a naive "always up" forecast.

With look-ahead, the model appears to explain more than 60% of the variance of 1- to 4-hour returns and to forecast their direction correctly in 76–80% of cases. With the causal decomposition, the R^2 is slightly negative at every horizon and the directional accuracy (49–51.5%) is no better than the naive "always up" benchmark.

4 Conclusions

1. Signal decompositions (VMD, EMD, wavelets) applied to a full series are *non-causal*: they must be recomputed on a rolling window of past data, both in training and in production.

2. A directional accuracy far above 50% on hourly crypto returns should be treated as a red flag for leakage rather than as evidence of skill.

3. Out-of-sample tests must be chronological, with a gap between periods, and compared with a simple benchmark (random walk, always-up).

4. In this case, the corrected model has no forecasting power: hourly Bitcoin returns behave essentially like a random walk at these horizons. The value of the exercise lies in the validation framework, which is directly transferable to model risk management.